Preston Crowe

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Security Engineer with 5+ years experience in detection engineering, threat intelligence infrastructure, and vulnerability research.

[EDUCATION]

University of Washington — Seattle, WA
Sep 2016 — Jun 2020
B.S. Computer Science, Dean's List, 3.71 GPA

Relevant Coursework: - Security · Cryptography · Networks · Hardware/Software Interface · Data Management · OS

[EXPERIENCE]

ExtraHop — Seattle, WA
Security Engineer III, December 2020 — Present

- Shipped detections within hours of disclosure for critical vulnerabilities and partnered with technical writers to publish expedited threat briefings.
- Built an Apache Iceberg indicator data lake serving analyst queries at 100M+ indicator scale; exported collections to a production enrichment API powering detections for all customers.
- Built the first ExtraHop MCP server for agentic enumeration of detections and devices, winning the 2025 hackathon audience choice award.
- Automated AWS traffic-mirroring environments to correlate detections against breach and attack simulation platforms; authored the internal Mandiant Security Validation guide used in customer engagements and added coverage for 3,000+ malicious file transfer actions.
- Developed detection strategies for red team tools (Redline, BloodHound, PetitPotam, Farmer); built pcap manipulation, device discovery, and automated traffic replay tooling with Threat Research and performance teams.
- Automated CVE monitoring and Jira ticket creation/triage for network vulnerabilities, eliminating tens of hours of manual researcher work per week; published daily CVE summaries and sentiment analysis.

Baffin Bay Networks — Seattle, WA
Internship, Product Development — June 2019 — September 2019

- Built an internal dashboard to monitor a honeypot network in React. Aggregated health metrics for sensors hosted across dozens of cloud providers distributed across the world.
- Developed an API for Baffin Bay's Threat Insight platform to provide IP context enrichment.

F5 Networks — Seattle, WA
Internship, F5 Labs — June 2018 — September 2018

- Worked with a team of security researchers to develop a vulnerability data pipeline.
- Built a threat map with React and D3.js to visualize suspicious traffic in real time.
- Generated data visualizations and animations for a Black Hat talk.

NASA AMES Research Center — Mountain View, CA
Internship, OpenMCT — June 2017 — September 2017

- Implemented JSON import/export functionality for workspaces and folder hierarchies in OpenMCT, an open-source mission control application.
- Worked in an enterprise environment and wrote extensive tests for each code contribution.

[SKILLS]

- **Detection & Response:** detection engineering, traffic analysis, incident response
- **Software Development:** Python, Go, JavaScript, Java, C, Git, SQL
- **System Architecture:** Linux, AWS, Docker, Terraform, Ansible
- **DevOps:** GitLab CI/CD, ETL, Chat Ops, Internal tools
- **Interests:** Proxmox, OPNSense, OpenWrt, SDR, ESP32, LoRa, OSINT

[PUBLICATIONS]

- F5 Networks: *Breaking Down the Door to Emergency Services through Cellular IoT Gateways*
- ExtraHop: *PetitPotam: Expanding NTLM Relay Attacks*